

Adatum - 19

Jamatul Ma'arif

Customer (cid: integer, ename: string, address: string, city: string, state: string)

Product (pid: integer, pname: string, price: integer, inventory: integer)

Shipment: (sid: integer, cid: integer, shipdate: Date/Time)

Shipped Product: (sid: integer, pid: integer, amount: integer)

① Return the product name and price for all products with a price less than \$5.00.

$\Rightarrow \pi(\text{pname}, \text{price}) (\sigma(\text{price} < 5.00) (\text{product}))$

② Return the id and names of all customers that have ever received a shipment where a product quantity (amount) in the shipment is greater than 20.

$\Rightarrow \pi(\text{cid}, \text{ename}) (\sigma(\text{sid} \in \pi(\text{sid}) (\sigma(\text{amount} > 20) (\text{shipment})))$

③ Return a list of all the customer names with address in the state 'CA' and have received a shipment prior to January 2, 2014.

$\Rightarrow \pi (ename) (\sigma (state = 'CA' \wedge shipdate < '2014-01-02'))$
 Customer $\bowtie$ Shipment

④ Return a list of product ids and names where the product has either a price $\geq \$25.00$ or has appeared in shipment with an amount greater than 10.

$\Rightarrow \pi (pid, pname) (\sigma (price \geq 25.00) (product)) \bowtie (\sigma (amount > 10) (shipment))$

ASUS X515
Spring - 20

Jannatul Maqsoom

Students: ID, s-name, address, mobile, email, gpa)

Course: (c-id, c-name, credit)

Department: (Dept-name, location)

Enrolled_by: sid, c-id, semester, year)

Offers: (c-id, Dept-name, semester, year)

① Find the student enrolled in the Spring 2019 semester.

$\Rightarrow \pi (s.id, s.name) \sigma (semester = 'Spring' \wedge year = 19) (Enrolled_by)$

② Find the courses offered under CSE department.

$\Rightarrow \pi (c-id, c-name) (\sigma (Dept_name = 'CSE') (Offers))$

③ Find the course by a particular student (ex: s_id = 'e101010')

$\Rightarrow \pi(c_id, c_name) (\sigma(s_id = 'e101010') \text{ Enrolled by } \bowtie \text{ course})$

④ Find the list of students enrolled in a course (cse-2423)

$\Rightarrow \pi(s_id, s_name) (\sigma(c_id = 'cse-2423') \text{ Enrolled by } \bowtie \text{ student})$

Autum-22

Jannatul Maaz

branch (branch_name, branch_city, assets)

customer (customer_name, customer_street,
customer_city)

amount (account_number, branch_name, balance)

loan (loan_number, branch_name, amount)

depositor (customer_name, account_number)

borrower (customer_name, loan_number)

① Find all customers who have a loan,
an account, or both.

$\Rightarrow \Pi \text{customer_name} (\text{depositor}) \cup \Pi \text{customer_name} (\text{borrower})$

② Find all customers who have a loan at the
Kumira branch.

$\Rightarrow \Pi \text{customer_name} (\text{branch_name} = \text{"Kumira"})$
and $\text{borrower.loan_number} = \text{loan.loan_number}$
(borrower X loan)

③ Find the names of all customers who have
a loan in GEE branch but do not have
an account in the bank.

$\Rightarrow \Pi \text{customer_name} (\text{branch_name} = \text{"GEE"} (\text{loan})) -$
 $\Pi \text{customer_name} (\text{branch_name} = \text{"GEE"} (\text{account}))$

④ find the account number of the millionaires of the bank.

$\Rightarrow \Pi \text{ account_name (6 balance)} = 10000000$ (acc)

Autum-18

Product (P-id, P-name, Unit-Price)

sales (s-id, P-id, e-id, Emp-id, Date-of-Sale, Time, Quantity)

customer: (c-id, c-name, Mobile, street, city, Country)

Employee: (Emp-id, E-name, Designation, Commission, Salary, Mobile, city, Country)

① find the Product sold to the customer

$\Rightarrow \Pi \text{ p-id, p-name, Unit-price (6 e-id = "100")}$
(sales X Product)

② find those Employee and their sales who have sold more than 10000 on 20-DEC-2018.

$\Rightarrow \Pi \text{ e-name, date-of-sale, time, quantity (6 quantity)}$
 $\Rightarrow \text{"10000"} \text{ and date-of-sales = "20-DEC-2018"}$
(Sales X Employee)

3. Find the customers details who have served by employee 'E'.
 $\Rightarrow \Pi e\text{-name, mobil, street, country, city} (G_{emp-id} = "E" (sales \times customer))$

④ Find the employees who have sold product 'p'.
 $\Rightarrow \Pi e\text{-name, designation, commission, salary, mobile, country, city} (G_{p-id} = "p" (sales \times Employee))$

⑤ Find the customer purchased product 'p' and from 'Dhaka'.
 $\Rightarrow \Pi e\text{-name, mobile, street, country, city} (G_{p-id} = "p" \text{ and } city = "DHAKA" (sales \times customer))$